

URBAN AIR QUALITY ANALYSIS: EXPLORING POLLUTION PATTERNS AND AQI PREDICTION ACROSS FIVE INDIAN CITIES

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DECLARATION

I hereby declare that the analysis, code, and findings presented in this report are my own original work, carried out as part of the IDRA Data Science & AI Training Program capstone project. All data cleaning, exploratory analysis, statistical checks, and model development were performed by me in the accompanying Jupyter notebook. Wherever external documentation, libraries, or references were used, they have been appropriately acknowledged in the References section. This report has not been copied from any other student's submission.

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ABSTRACT

This report analyses a ten-year daily air-quality file covering Bangalore, Chennai, Delhi, Kolkata and Mumbai, supplied as the dataset for Project 8 (P_8_city_day.csv, 18,265 rows and 16 columns). The aim was to describe how the twelve recorded pollutants and the reported Air Quality Index vary by city and over time, and to test whether those pollutants can estimate AQI through regression. The file contained no missing values and no duplicate rows, so cleaning concentrated on verifying data types, date coverage and internal consistency rather than repairing damaged records. The exploratory stage produced an unexpected result: median AQI differs by less than 20 points across the six severity labels, and every pollutant correlates with AQI at 0.02 or below under both Pearson and Spearman measures. Two further checks confirmed the pattern: twelve of the thirteen numeric columns tested are statistically indistinguishable from a uniform draw over their own range, and the supplied labels agree with the published CPCB bands for only 16.2 per cent of rows, close to the 16.7 per cent expected by chance. Five regression variants were compared on a chronological split, training on 2015-2022 and testing on 2023-2024. None improved on the mean-baseline MAE of 125.14 AQI units. Random Forest reached a training R2 of 0.82 but fell to -0.02 on the test period, a clear case of fitting noise. The conclusion is that this file carries no usable relationship between pollutants and AQI, and the report presents that null result rather than forcing a positive one.

Keywords: exploratory data analysis; data validation; regression; AQI prediction; correlation analysis; overfitting

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1. INTRODUCTION

1.1 Background

Air pollution is one of the more visible public-health problems in Indian cities. Vehicle traffic, industrial activity, construction dust and seasonal weather all push pollutant concentrations up and down over the course of a year. The Air Quality Index is the single number used to communicate how bad the air is on a given day, and it is normally worked out from sub-indices for individual pollutants such as PM2.5, PM10, NO2, SO2, CO and O3. How AQI relates to those underlying measurements, and whether that relationship can be modelled from data, is therefore a useful question to put to a dataset.

1.2 Context of the Problem

The file used here was supplied for Project 8 of the IDRA capstone, "Urban Air Quality Analysis: Understanding Environmental Drivers and Predicting Air Quality". It records twelve pollutant measurements alongside a numeric AQI value and a categorical severity label for five cities, Bangalore, Chennai, Delhi, Kolkata and Mumbai, on every day from 1 January 2015 to 31 December 2024. City planners, pollution-control boards and public-health researchers are the obvious audience for this kind of work. If pollutant readings reliably predict AQI, that relationship could support monitoring, early warning or decisions about where to direct resources.

1.3 Scope of the Study

This project works only with the supplied CSV. It covers structural and quality checks on the data, exploratory and statistical analysis of pollutant and AQI patterns across cities and time, a regression exercise that attempts to estimate AQI from the twelve pollutants plus city and simple calendar features, and an evaluation of how well that regression performs, including model comparison, overfitting checks and error analysis. It does not verify the file against an official government source, does not bring in weather or traffic data, and does not attempt causal inference. As Sections 8 to 13 show, the analysis also turned up a serious internal inconsistency in the data, since the supplied AQI category labels do not track the numeric AQI column. That inconsistency limits how far any modelling conclusion can be taken.

2. PROBLEM STATEMENT AND MOTIVATION

The main question is whether the twelve pollutant measurements in this file, together with city identity and calendar information, can estimate the numeric AQI recorded on the same row. A second question came out of the validation work described in Section 6: are the supplied category labels (Good, Satisfactory, Moderate, Poor, Very Poor, Severe) themselves consistent with the numeric AQI column? Both matter. A regression model is only worth having if it beats a trivial baseline, and a category label is only worth having if it reflects the measurement it claims to summarise. This project treats both as questions to be tested rather than assumptions to be accepted.

3. OBJECTIVES

- To inspect and validate the structure, data types, and completeness of the supplied air-quality dataset.
- To characterise how pollutant concentrations and AQI vary across the five cities and over the ten-year period.
- To check whether the supplied AQI category labels are consistent with the numeric AQI values.
- To quantify the statistical association between each pollutant and AQI using Pearson and Spearman correlation.

- To build and compare regression models (mean baseline, Linear Regression, Random Forest, with and without calendar features) that estimate AQI from pollutant measurements, using a chronological train-test split.
- To evaluate model performance carefully, including overfitting and underfitting checks and error analysis by city and by AQI range, and to turn the results into clear conclusions and recommendations.

4. RESEARCH QUESTIONS

1. What is the structure and quality of the supplied dataset (rows, columns, missing values, duplicates, data types)?
2. How do pollutant measurements and AQI vary across the five cities and across calendar time?
3. Are the supplied AQI category labels consistent with the numeric AQI values recorded in the same rows?
4. Which pollutant variables show the strongest observed association with AQI?
5. How accurately can a regression model estimate AQI from the available pollutant, city and calendar features, and does the model generalise from earlier to later dates?
6. Where do prediction errors concentrate, and what does the error pattern imply about the model and the data?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset (P_8_city_day.csv) was supplied by IDRA through the LMS as the assigned file for Project 8, "Urban Air Quality Analysis". It is a city-day panel of air-quality measurements; no external license terms accompanied the file, and no additional government or third-party source was merged into it for this project.

5.2 Dataset Size and Structure

The file contains 18,265 rows and 16 columns. It covers five cities, Bangalore, Chennai, Delhi, Kolkata and Mumbai, each with exactly 3,653 daily records spanning 1 January 2015 to 31 December 2024. There are no gaps in the date coverage and no duplicated city-date pairs. The target for the regression task is AQI, a continuous numeric field. A parallel categorical field, AQI_Bucket, groups AQI into six labels, but it was kept out of the model inputs because it is derived from the target itself. Section 7 explains that decision.

Property	Value
Rows (observations)	18,265
Columns (variables)	16
Cities covered	Bangalore, Chennai, Delhi, Kolkata, Mumbai (5)
Date range	2015-01-01 to 2024-12-31 (daily, 10 years)
Records per city	3,653 (identical for every city)

Property	Value
Target variable	AQI (continuous, regression)
Missing values	0 (after date/numeric parsing)
Exact duplicate rows	0

Table 1. Summary of the Dataset

5.3 Data Dictionary

Column	Meaning	Type	Role
City	Monitoring city	Categorical (5 levels)	Feature
Datetime	Observation date	Date	Feature (calendar) / index
PM2.5, PM10	Particulate matter concentration ($\mu\text{g}/\text{m}^3$)	Numeric	Feature
NO, NO2, NOx	Nitrogen oxide compounds ($\mu\text{g}/\text{m}^3$)	Numeric	Feature
NH3, CO, SO2, O3	Ammonia, carbon monoxide, sulphur dioxide, ozone ($\mu\text{g}/\text{m}^3$ or mg/m^3)	Numeric	Feature
Benzene, Toluene, Xylene	Volatile organic compounds ($\mu\text{g}/\text{m}^3$)	Numeric	Feature
AQI	Composite Air Quality Index	Numeric (continuous)	Target
AQI_Bucket	Categorical AQI severity label	Categorical (6 levels)	Excluded (target leakage)

Table 2. Data Dictionary

6. DATA CLEANING

By the standards of real air-quality files, this dataset arrived clean. There were no missing values and no duplicate rows once dates and numeric fields had been parsed. The cleaning stage therefore became a matter of confirming that this was actually true rather than assuming it, and of looking for the subtler problems described below.

6.1 Missing Values

All 16 columns had zero missing values before and after converting the `Datetime` column to a date format and the other non-categorical columns to numeric values. Since no new missing values appeared during conversion, the values could be parsed successfully using these checks.

6.2 Duplicate Records

No exact duplicate rows were found, since `df.duplicated().sum()` returned 0. A stricter check for duplicate City-Datetime pairs, which would mean two conflicting readings for the same city on the same day, also returned 0. No rows were removed.

6.3 Data Types and Inconsistent Values

City and AQI_Bucket were retained as text/categorical columns; all other columns were coerced to numeric and Datetime to a proper date type. Each city showed exactly 3,653 unique dates with no gaps, confirming a complete daily panel for all five cities over the ten-year window.

6.4 Outlier Handling

No automatic outlier rule was applied. Instead the summary statistics and a count of exact-zero readings per column were examined first. Every pollutant and the AQI column runs from 0 up to a fixed ceiling, for example 0 to 499.9 for PM2.5, 0 to 600 for PM10 and 0 to 500 for AQI. In each case the mean and median sit close to the midpoint of that range and the standard deviation is roughly half the range. That is the pattern you would expect from values spread evenly across a fixed interval, not the strongly right-skewed distributions that real monitoring data produces. Since no value fell outside a plausible physical range and there was no extreme tail to deal with, nothing was capped or removed. The flat shape is reported as a data-quality observation in Section 15 rather than treated as a defect to be corrected.

Column	Mean	Std	Min	Median	Max	Exact zeros
PM2.5	250.60	144.46	0.00	251.00	499.90	1
PM10	299.44	173.48	0.00	300.30	600.00	2
NO2	75.42	43.46	0.00	76.00	150.00	7
CO	5.00	2.89	0.00	5.00	10.00	11
SO2	49.84	28.99	0.00	49.90	100.00	10
O3	100.41	57.59	0.00	100.70	200.00	6
AQI	251.11	144.50	0.00	251.20	500.00	1

Table 3. Selected Descriptive Statistics and Zero-Value Counts (see notebook for the full twelve-pollutant table)

6.5 Before vs After Cleaning

Check	Before	After
Missing values (all columns)	Unknown until parsed	0
Duplicate rows	0	0 (confirmed)
Duplicate City-Datetime pairs	Unknown until checked	0

Check	Before	After
Datetime column type	Text (object)	Parsed datetime, no invalid dates
Numeric columns	Read generically by pandas	Explicitly coerced, 0 new NaNs introduced

Table 4. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

7.1 Categorical Encoding

City is the only categorical predictor retained for modelling. It was one-hot encoded (OneHotEncoder with `handle_unknown="ignore"`) inside a scikit-learn ColumnTransformer, so that unseen category values in the test period would not break the pipeline. AQI_Bucket was deliberately excluded from every model's feature set because it is derived from the same AQI value the model is trying to predict; including it would be target leakage.

7.2 Feature Scaling and Transformation

The twelve pollutant columns were median-imputed and standardised with StandardScaler before being passed to Linear Regression, since linear models are sensitive to feature scale. Random Forest does not need scaling, but the same pipeline was reused for both so that the comparison stayed fair and the code stayed simple. All imputation and scaling parameters were fitted on the training partition only and then applied to the test partition, which prevents information from the test period leaking into the preprocessing statistics.

7.3 Train-Test Split

The data is a daily time series, so a random row-wise split would have let the model learn from dates that come after the ones it is tested on. A chronological split was used instead. The earliest 80 per cent of unique dates form the training set and the latest 20 per cent form the test set. That gives a training period of 1 January 2015 to 31 December 2022 with 14,610 rows, a test period of 1 January 2023 to 31 December 2024 with 3,655 rows, and a cutoff of 1 January 2023. This setup asks the more realistic question of whether patterns learned on earlier data hold up on later, unseen dates.

7.4 Final Feature Set

Baseline feature set: the twelve pollutant measurements (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene) plus one-hot encoded City. Calendar-enhanced feature set: the same twelve pollutants and City, plus `month_sin`, `month_cos` (cyclical month-of-year encodings) and year. Target variable: AQI.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

The twelve pollutant measurements were plotted as histograms in a grid of small multiples. Real pollutant readings usually produce a peaked, right-skewed shape. These do not. Each histogram is close to flat across its observed range, which matches the pattern already noted in Section 6.4.

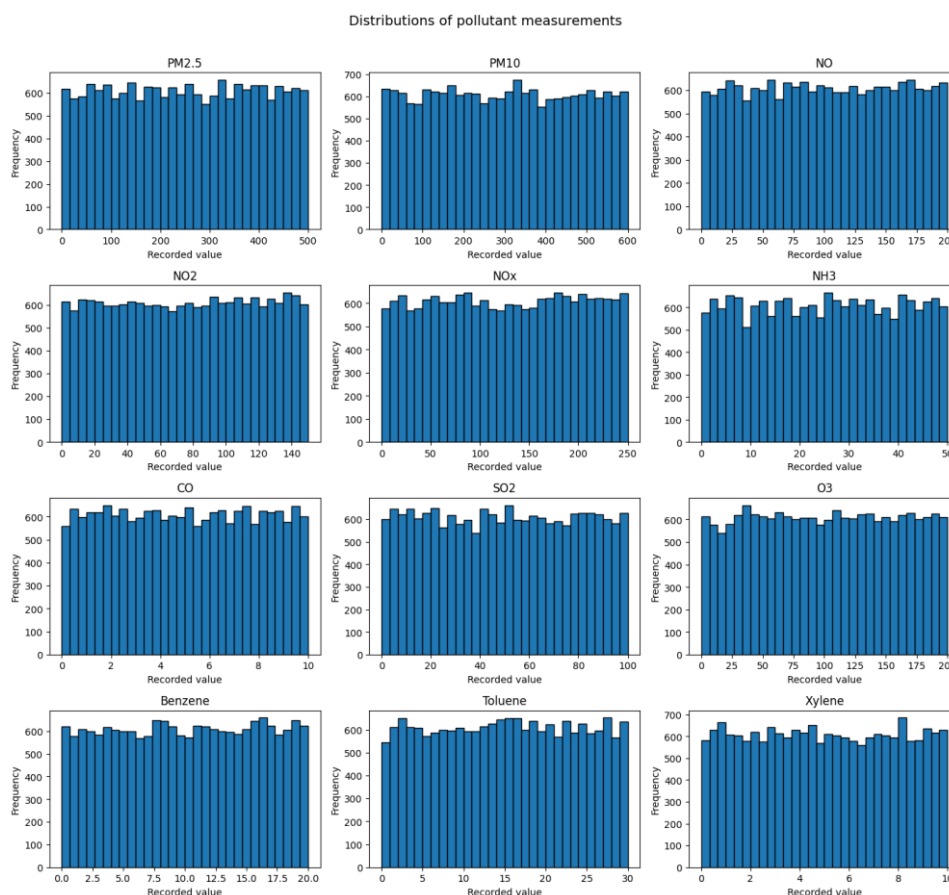


Figure 1. Distributions of the twelve Pollutant Measurements

Every pollutant produces an approximately flat histogram across its full observed range instead of the right-skewed shape expected of real pollution readings. This supports the data-quality observation made in Section 6.

8.2 Categorical Analysis

Mean AQI was compared across the five cities. The means sit between about 249.8 for Bangalore and 253.3 for Mumbai, a spread of under four points on a variable whose theoretical range is 0 to 500. In practical terms there is no difference in average AQI between cities in this file.

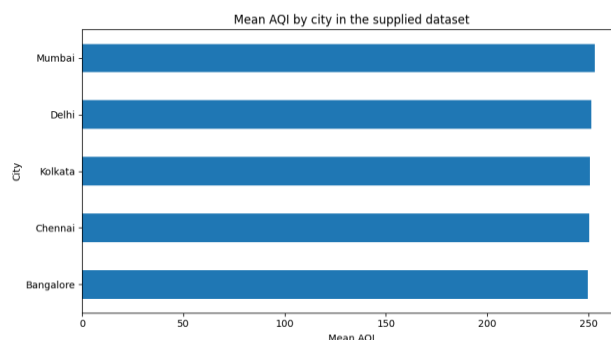


Figure 2. Mean AQI by City

City does not appear to be a strong differentiator of AQI in this dataset: all five cities average close to 250, near the midpoint of the AQI scale.

8.3 Bivariate / Relationship Analysis

AQI was tracked over time (monthly means per city) to look for trends or seasonality that a real air-quality series would typically show (for example, higher AQI in the dry winter months in northern India).

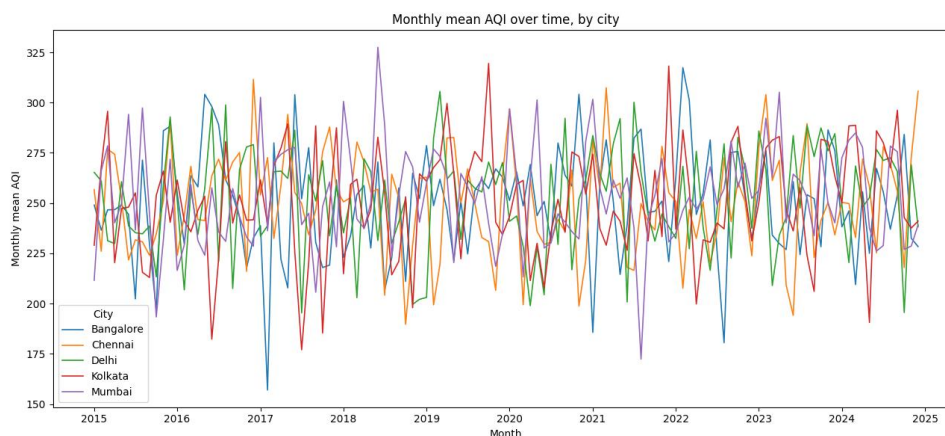


Figure 3. Monthly Mean AQI Over Time, by City

The monthly series oscillates around 250 for all five cities with no visible long-run trend and no consistent seasonal cycle across the ten years. Indian cities normally show a pronounced winter peak, particularly in the north, so its absence here is another sign that this file does not reproduce real seasonal behaviour.

8.4 Correlation Analysis

Pearson correlation was computed between every pollutant and AQI, then shown two ways: as a ranked bar chart (Figure 4) and as a full correlation matrix (Figure 5).

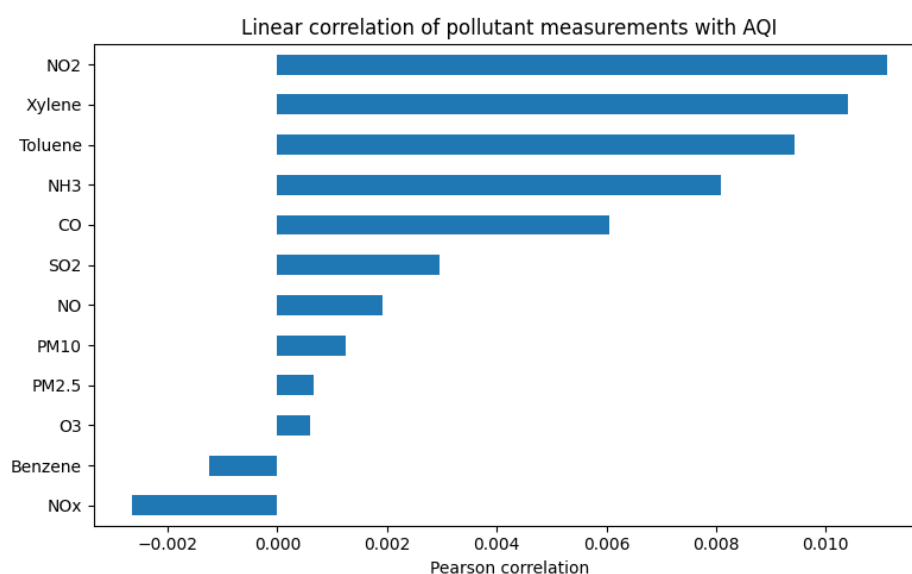


Figure 4. Linear Correlation of Each Pollutant with AQI

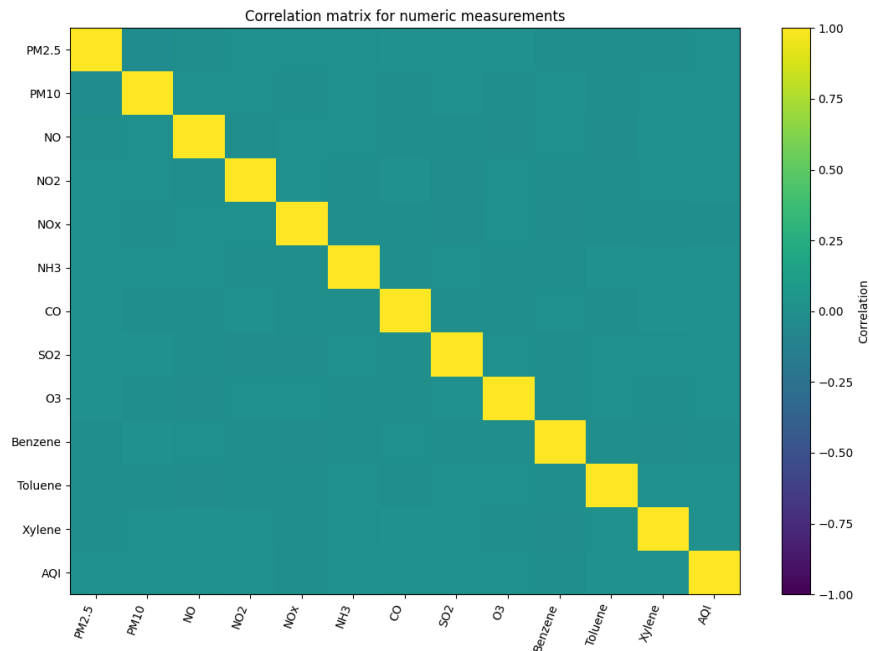


Figure 5. Correlation Matrix for All Numeric Variables

Every pollutant-AQI correlation falls between -0.02 and 0.01 in magnitude, and the pollutant-to-pollutant correlations are similarly close to zero. This matters a great deal. In a real pollution dataset PM2.5, PM10 and the nitrogen-oxide family are usually strongly correlated with each other and strongly associated with AQI, for the simple reason that AQI is calculated from those very measurements. Finding none of that structure suggests either that the AQI column in this file was not derived from the accompanying pollutant columns, or that the columns were generated independently of one another.

8.5 AQI Category Label Consistency

AQI_Bucket is supposed to summarise AQI severity, so the numeric AQI distribution within each supplied label was examined directly in Figure 6.

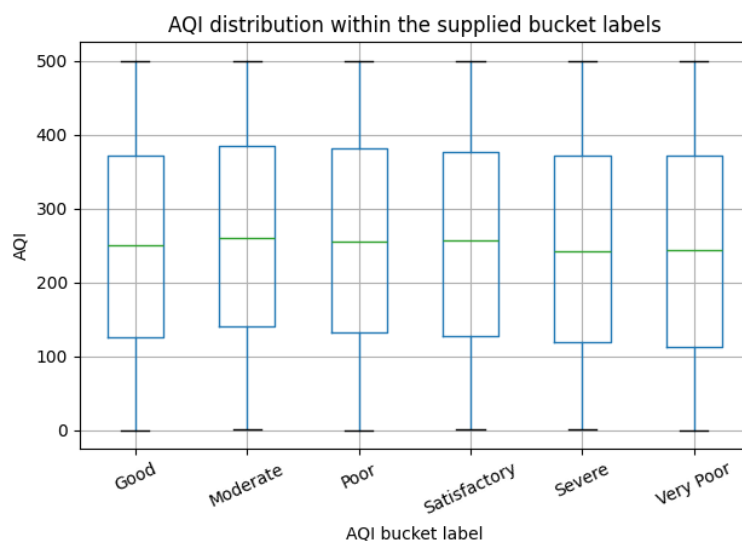


Figure 6. AQI Distribution Within the Supplied AQI_Bucket Labels

If the labels summarised AQI severity correctly, "Good" would show the lowest AQI values and "Severe" the highest. They do not. The medians are Good 249.9, Satisfactory 257.1, Moderate 260.1, Poor 255.0, Very Poor 243.9 and Severe 241.2, all clustered within a 19-point band around the dataset median of 251.2, with interquartile ranges that overlap almost completely. The labels in this file do not track the

numeric AQI they are meant to describe, so AQI_Bucket cannot be used as a severity indicator and was rightly excluded from the model.

A second, more direct test was then run. The Central Pollution Control Board publishes fixed AQI bands: 0 to 50 Good, 51 to 100 Satisfactory, 101 to 200 Moderate, 201 to 300 Poor, 301 to 400 Very Poor and 401 to 500 Severe. Cross-tabulating the supplied labels against the band implied by each row's numeric AQI shows that the two agree for only 16.2 per cent of rows. Assigning one of six labels at random would give about 16.7 per cent. Rows labelled "Severe" are spread almost evenly across the whole 0 to 500 range, with 342 of them falling in the 0 to 50 band. The labels in this file were therefore not produced from the numeric AQI column at all.

9. STATISTICAL ANALYSIS

City-level descriptive statistics (mean, median, standard deviation, quartiles and IQR) confirm the near-identical AQI distributions noted in Section 8.2: all five cities show a mean near 250, a standard deviation near 144–145, and an IQR of roughly 249–254 AQI points.

City	Mean	Median	Std	Q1	Q3	IQR
Mumbai	253.32	254.40	145.38	126.50	380.00	253.50
Delhi	251.50	257.20	143.67	125.10	373.30	248.20
Kolkata	250.64	249.50	144.29	125.40	376.70	251.30
Chennai	250.34	248.60	144.90	125.00	375.60	250.60
Bangalore	249.77	247.80	144.34	125.50	376.40	250.90

Table 5. Descriptive Statistics of AQI by City

Spearman rank correlation was used alongside Pearson because it picks up monotonic relationships that are not straight lines, and because it is less affected by the shape of the underlying distribution. For all twelve pollutants Spearman's rho is at most 0.01 in absolute value, and every p-value exceeds 0.05. The smallest, for NO₂, is 0.14. None of the pollutant-AQI associations is statistically significant, and that is despite a sample of 18,265 rows, which is large enough that even a weak genuine effect would normally show up. The result lines up with the near-zero Pearson correlations in Section 8.4: there is no reliable monotonic or linear association between any single pollutant and AQI in this file.

Pollutant	Spearman rho	p-value
NO ₂	0.01	0.14
Xylene	0.01	0.16
Toluene	0.01	0.20
NH ₃	0.01	0.28
CO	0.01	0.41

Pollutant	Spearman rho	p-value
SO2	0.00	0.69
Others (NOx, NO, Benzene, PM10, PM2.5, O3)	≤ 0.00	0.73 – 0.94

Table 6. Spearman Correlation of Pollutants with AQI (n = 18,265 for each pair)

A further check was run to characterise the columns directly rather than only to show that relationships are missing between them. A one-sample Kolmogorov-Smirnov test compared each of the thirteen numeric columns, all twelve pollutants plus AQI, against a uniform distribution over its own observed range. Real pollutant readings are strongly right-skewed and would reject uniformity easily. Twelve of the thirteen columns do not: p-values range from 0.11 to 0.97, and uniformity cannot be rejected for any of them at the 5% level. The one exception is NOx, which rejects uniformity at $p = 0.022$. With thirteen independent tests at that threshold, roughly 0.65 false rejections are expected by chance alone, so this single borderline result does not overturn the general pattern, though it is reported rather than set aside. Combined with the near-zero correlations and the 16.2 per cent label agreement reported in Section 8.5, the pattern is consistent with values drawn independently and close to uniformly rather than measured, though the file alone cannot confirm how it was produced.

10. FEATURE ENGINEERING

Two calendar-derived features were engineered from the Datetime column, in addition to the raw pollutant measurements and city:

Feature	Formula / Logic	Rationale
month_sin, month_cos	$\sin(2\pi \cdot \text{month}/12)$, $\cos(2\pi \cdot \text{month}/12)$	Represents month cyclically so December and January are treated as adjacent, avoiding an artificial jump at year-end that a raw month number (1–12) would create.
year	Calendar year of the observation	Allows the model to capture any long-run drift across the ten-year span.
day_of_year	Day number within the year (1–366)	Computed as an intermediate step for the cyclical encodings; not used directly as a model input.

Table 7. Engineered Calendar Features

These features record when an observation was taken. They do not show that the calendar period caused the AQI value. Section 8.3 found no visible seasonal cycle, so the calendar features were expected to add little, and Section 12 confirms that they did. Including them provides a fair test of the expectation rather than an assumption about it.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

This is a regression task: the target, AQI, is a continuous numeric variable. The features are the twelve pollutant measurements, the one-hot encoded city, and (in the calendar-enhanced variant) the month_sin, month_cos and year features. AQI_Bucket is excluded as a feature because it is derived from the target.

11.2 Model Selection

Three model families were compared:

- Mean baseline (DummyRegressor, strategy="mean"): predicts the training-set mean AQI for every test row. Any model that cannot beat this baseline is not adding predictive value.
- Linear Regression: a simple, interpretable model that assumes an additive linear relationship between the (scaled) pollutant features, city, and AQI.
- Random Forest Regressor (150–200 trees, min_samples_leaf = 2): a non-linear ensemble model capable of capturing interactions and non-linear relationships, used here as a more flexible alternative to Linear Regression.

Each model was trained twice: once on the twelve pollutants + city ("baseline feature set"), and once on the same features plus the three calendar features ("calendar-enhanced feature set"), using the identical chronological train/test split described in Section 7.3.

11.3 Training the Model

All models were trained inside a scikit-learn Pipeline combining a ColumnTransformer (median imputation + standard scaling for numeric features; most-frequent imputation + one-hot encoding for City) with the regressor, so preprocessing statistics were fit only on the training partition. The key training code is reproduced below; the full pipeline is available in the accompanying notebook and Appendix A.

```
preprocessor = ColumnTransformer([
    ("numeric", Pipeline([("imputer", SimpleImputer(strategy="median")),
                          ("scale", StandardScaler())]), numeric_features),
    ("city", Pipeline([("imputer", SimpleImputer(strategy="most_frequent")),
                      ("onehot", OneHotEncoder(handle_unknown="ignore"))]), ["City"]),
])

models = {
    "Mean baseline": DummyRegressor(strategy="mean"),
    "Linear Regression": Pipeline([("prep", preprocessor), ("model",
LinearRegression())]),
    "Random Forest": Pipeline([("prep", preprocessor),
                              ("model", RandomForestRegressor(n_estimators=150,
min_samples_leaf=2, random_state=42, n_jobs=-1))]),
}
for name, model in models.items():
    model.fit(X_train, y_train)
    predictions[name] = model.predict(X_test)
```

12. MODEL EVALUATION

12.1 Regression Evaluation

Models were evaluated on the held-out 2023–2024 test period using Mean Absolute Error (MAE, average absolute prediction error in AQI units), Root Mean Squared Error (RMSE, penalises larger errors more heavily) and R^2 (share of test-set AQI variance explained relative to the test-set mean; negative values mean the model performs worse than always predicting that mean).

Model	MAE	RMSE	R^2
Mean baseline	125.14	144.02	-0.00
Linear Regression	125.15	144.03	-0.00
Random Forest	126.17	145.44	-0.02
Linear Regression + calendar	125.08	143.97	-0.00
Random Forest + calendar	125.96	145.37	-0.02

Table 8. Regression Model Performance on the 2023–2024 Test Period

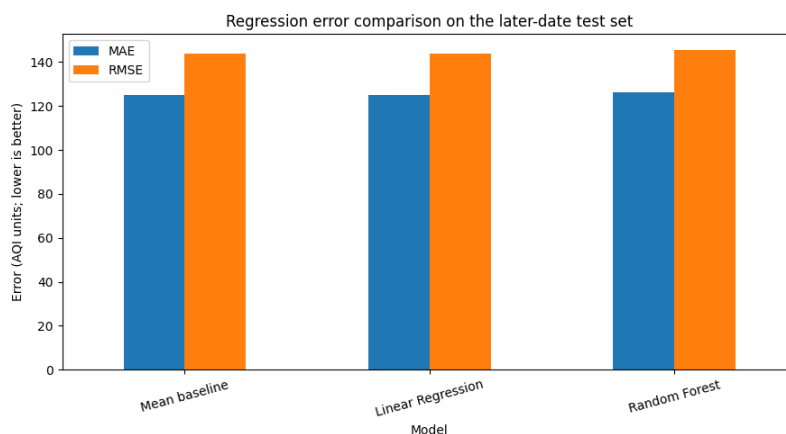


Figure 7. Regression Error Comparison (MAE and RMSE) for the Baseline Feature-Set Models

No model meaningfully beats the mean baseline. MAE and RMSE are essentially flat across all five variants and every R^2 value sits at or below zero. Adding calendar features gives Linear Regression a 0.07-unit improvement in MAE and gives Random Forest nothing. Given the near-zero correlations reported in Sections 8.4 and 9, this is the expected outcome. No model can extract a relationship from data that shows no statistical association in the first place.

12.2 Overfitting and Underfitting

Training and test performance were compared for all five variants to see whether any model was overfitting, meaning it had fitted noise in the training data that does not carry over, or underfitting, meaning it had failed to capture signal available on either partition.

Model	Train MAE	Test MAE	Train R^2	Test R^2
Mean baseline	125.15	125.14	0.00	-0.00

Model	Train MAE	Test MAE	Train R ²	Test R ²
Linear Regression	125.09	125.15	0.00	-0.00
Random Forest	52.45	126.17	0.82	-0.02
Linear Regression + calendar	125.05	125.08	0.00	-0.00
Random Forest + calendar	52.13	125.96	0.82	-0.02

Table 9. Training vs. Test Performance for All Five Model Variants

Random Forest shows the classic signs of overfitting. Its training R² of 0.82, from a training MAE of only 52.45, falls to a test R² of -0.02 and a test MAE of 126.17, a gap of about 74 MAE units between the two partitions. The calendar-enhanced version behaves identically, with a training R² of 0.82 and a test R² of -0.02. Because the underlying pollutant-AQI relationship is absent, the flexible model has nothing genuine to learn and instead memorises noise specific to the training rows, so its held-out performance drops back to slightly worse than the trivial baseline. Linear Regression shows almost identical training and test error, though this is not evidence that it generalises well. A linear model with no linear signal to fit simply converges on predicting close to the training mean, which is what the baseline does anyway. None of the models is underfitting in the usual sense of being too simple to capture available signal. The signal itself appears to be missing from the file.

13. RESULTS AND FINDINGS

1. The dataset is complete, with no missing values and no duplicates, and structurally regular, with exactly 3,653 daily records per city over identical date ranges. However, twelve of the thirteen numeric columns tested, all pollutants plus AQI, are distributed almost uniformly across a fixed range (Figure 1, Table 3), unlike the right-skewed shapes that real air-quality readings produce. A Kolmogorov-Smirnov test cannot reject uniformity for any of these at the 5 per cent level except NO_x ($p = 0.022$), a borderline result not unusual given thirteen tests were run.
2. The supplied AQI_Bucket labels are not consistent with the numeric AQI column. Median AQI across the six labels spans only 241.2 to 260.1 (Figure 6), and the labels match the published CPCB band implied by the numeric AQI for just 16.2 per cent of rows, against 16.7 per cent expected from random assignment. The field cannot be used as a severity indicator and was correctly kept out of the model.
3. City makes almost no difference to AQI in this file. Mean AQI across the five cities varies by under four points on a 0 to 500 scale (Figure 2, Table 5).
4. No pollutant shows a meaningful association with AQI, statistically or practically. All Pearson correlations fall between -0.02 and 0.01 (Figures 4 and 5), and every Spearman p -value exceeds 0.05 (Table 6) despite a sample of more than 18,000 rows.
5. None of the five regression variants improves meaningfully on the mean-baseline MAE of 125.14 AQI units, and the best test R² achieved is -0.00 (Table 8, Figure 7).
6. Both Random Forest variants overfit severely, with a training R² of 0.82 against a test R² of -0.02 (Table 9). Error analysis shows the errors are largest at the extremes of the AQI range and roughly flat across cities (Table 10), which is what happens when a model is effectively predicting a near-constant value because there is no real signal to follow.

14. DISCUSSION

The exploratory work, the statistical checks and the modelling results all point the same way. Pollutant-AQI correlations are close to zero under both Pearson and Spearman measures, AQI barely varies by city or by calendar month, and the supplied category labels agree with the published CPCB bands at roughly the rate expected from random assignment. Twelve of the thirteen numeric columns tested are also statistically indistinguishable from a uniform draw over their own range; NO_x is a borderline exception ($p = 0.022$), though a single such result among thirteen tests at the 5 per cent level is not unusual by chance. Taken together these say that within this file, AQI does not appear to have been derived from, or to co-vary meaningfully with, the accompanying pollutant measurements, and the category labels do not appear to correspond to the numeric AQI recorded in the same rows. That explains the modelling results rather than sitting awkwardly beside them, because a model cannot learn a relationship the data does not contain. The finding that nothing beats a mean baseline is itself evidence for the same conclusion.

The Random Forest result is worth dwelling on for what it demonstrates about method. Any model with enough capacity can drive training error down, even on data with no exploitable structure, simply by memorising incidental patterns in the training rows. The chronological holdout and the train-versus-test comparison in Table 9 exist precisely to expose that. A training R² of 0.82 reported on its own would have looked like a successful model, and only the test R² of -0.02 shows what it really is. This is a concrete illustration of why the capstone guide asks for both figures rather than training performance alone.

It is equally important to be clear about what this project does not show. Finding no pollutant-AQI relationship in this particular CSV says nothing about the real world, where the relationship is well established, since AQI is normally computed directly from pollutant sub-indices. The defensible reading is that this file does not preserve that relationship. What produced the file, whether synthetic generation, anonymisation, scrambling or some other process upstream of this project, cannot be determined from the file alone. The task here was to find and report the inconsistency honestly, not to speculate about its cause or manufacture a positive result.

15. LIMITATIONS

- Undocumented source. The CSV was supplied for the assignment without documentation of the original measurement stations, instruments, collection process, or the formula used to calculate AQI, making it impossible to verify why the pollutant-AQI relationship is absent.
- Distributional anomaly. Twelve of the thirteen numeric columns tested, all pollutants plus AQI, are consistent with a uniform draw over a fixed range, which is unusual for real air-quality data. This raises questions about how the file was produced, but does not by itself confirm whether the values were measured, synthetically generated, or altered in some other way; that would need to be verified with the data provider.
- Label inconsistency. Because AQI_Bucket does not track the numeric AQI (Section 8.5), any downstream use of the category field in this file would be unreliable.
- Single evaluation split. The chronological 80/20 holdout gives one assessment of generalisation. A different cutoff date, or a rolling-origin evaluation, could give a different answer, although given the near-zero correlations a materially different result seems unlikely.
- Model scope. Only a mean baseline, Linear Regression and Random Forest were tested. Other families such as gradient boosting or regularised linear models were not explored, though with no linear or rank correlation present they would be unlikely to change the conclusion.
- No causal claims. Correlations and feature usage describe association only. No causal relationship between any pollutant and AQI can be inferred here, and that would remain true even if the correlations had been substantial.

- Limited coverage. The file covers five named Indian cities over one ten-year window. The findings should not be extended to other cities, regions or periods without independent verification.

16. CONCLUSION

This project set out to understand how air-quality measurements vary across five Indian cities and to test whether the supplied pollutant measurements could estimate AQI through regression. The dataset was structurally clean, with no missing values, no duplicates and complete daily coverage, which meant the analysis could move quickly to exploratory and statistical work. That work found almost no variation in AQI by city or by season, no meaningful linear or rank correlation with any of the twelve pollutants, and category labels inconsistent with the numeric AQI they are supposed to summarise. Two further checks reinforced the picture: twelve of the thirteen numeric columns tested cannot be distinguished from uniform draws over their own range, and the labels match the published CPCB bands for only 16.2 per cent of rows. Consistent with all of this, none of the five regression variants improved on a naive mean-baseline prediction over a chronologically held-out test period, and the Random Forest's overfitting confirmed that its apparent training fit reflected nothing generalisable. The project concludes that within this file, pollutant measurements cannot be used to estimate AQI reliably, and that this null result is the accurate outcome of the analysis rather than a failure of it.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

- Confirm where the data came from before using it again. The pollutant columns show no association with AQI and the labels do not match the published bands, so the source, collection method and AQI calculation procedure should be checked with IDRA or the original provider before this file is used for teaching or any decision beyond this exercise.
- Do not treat AQI_Bucket as a severity indicator for this file. The category medians overlap almost completely and the labels match the CPCB bands only 16.2 per cent of the time (Section 8.5), so any report or dashboard built on it should either recompute the categories from the numeric AQI using the documented thresholds or flag the supplied labels as unreliable.
- Do not put these models into operational use. Since none of them beats the mean baseline, using any of them to predict AQI in a real monitoring context would add nothing over simply reporting the historical average.
- Use a differently constructed dataset if genuine forecasting is the goal. That would mean documented, independently sourced pollutant and AQI data, ideally alongside weather and traffic covariates, and lagged predictors that would actually be available before the forecast time rather than same-day measurements.
- Test stability across several evaluation windows. On a verified dataset, model comparison should be repeated across multiple chronological cutoffs rather than a single split, so that any positive result has been shown to hold before it is trusted.

17.2 Future Work

There are four obvious ways to take this further. The first is to obtain a verified, documented air-quality dataset, for example from the Central Pollution Control Board, and repeat the analysis on it as a comparison against the supplied file. The second is to add weather variables such as temperature, humidity and wind speed, along with proxies for traffic or industrial activity. The third is to replace the single chronological split with rolling-origin cross-validation. The fourth is to benchmark any future model against a persistence baseline, using yesterday's AQI as today's forecast, since persistence is usually a tougher reference than the mean for environmental time series that are genuinely autocorrelated.

18. REFERENCES

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Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.

Virtanen, P., et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. Nature Methods, 17, 261-272.

APPENDIX A. CODE

A.1 Data Loading and Inspection

```
df = pd.read_csv("P_8_city_day.csv")
print("Dataset shape:", df.shape)          # (18265, 16)
print(df.isna().sum())                     # 0 missing in every column
print("Exact duplicate rows:", df.duplicated().sum()) # 0
```

A.2 Data Cleaning

```
df["Datetime"] = pd.to_datetime(df["Datetime"], errors="coerce")
for col in df.columns:
    if col not in ["City", "AQI_Bucket", "Datetime"]:
        df[col] = pd.to_numeric(df[col], errors="coerce")
```

A.3 Preprocessing and Chronological Split

```
unique_dates = np.array(sorted(model_data["Datetime"].unique()))
split_index = int(len(unique_dates) * 0.8)
cutoff_date = pd.Timestamp(unique_dates[split_index])
train_df = model_data[model_data["Datetime"] < cutoff_date]
test_df = model_data[model_data["Datetime"] >= cutoff_date]
```

A.4 EDA and Statistical Checks

```
from scipy.stats import spearmanr
for col in pollutants:
    rho, p_value = spearmanr(df[col], df["AQI"])
```

A.5 Model Training and Evaluation

```
for name, fitted in fitted_models.items():
    train_pred = fitted.predict(X_train)
```

```
test_pred = predictions[name]
# Train_MAE, Test_MAE, Train_R2, Test_R2 computed for overfitting check
```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

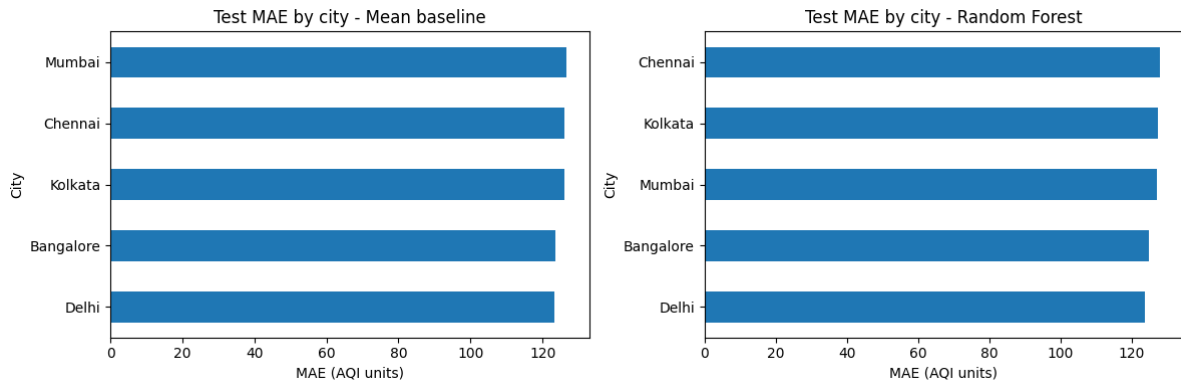


Figure 8. Test-Set MAE by City, Mean Baseline and Random Forest

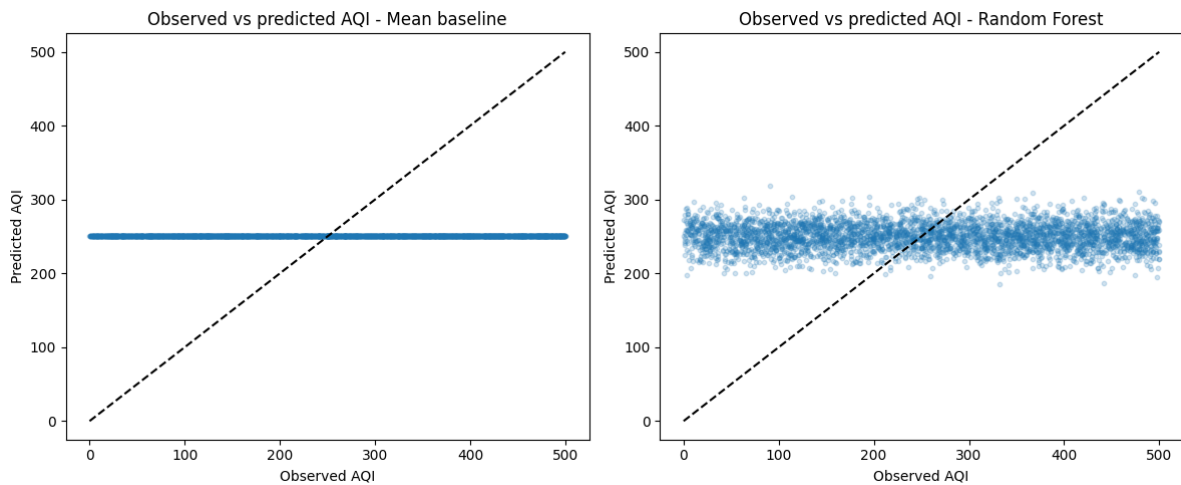


Figure 9. Observed vs. Predicted AQI on the Test Set, Mean Baseline and Random Forest

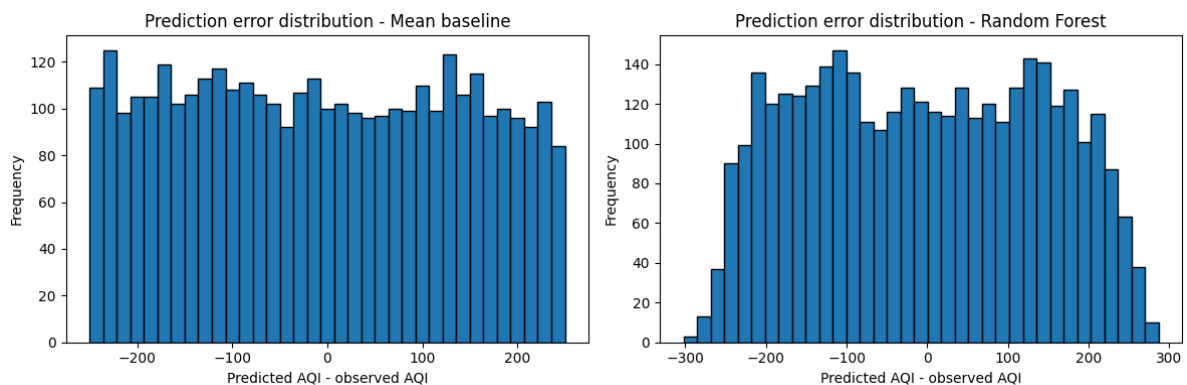


Figure 10. Prediction Error Distribution on the Test Set, Mean Baseline and Random Forest

AQI quartile (test set)	Records	MAE	Mean error
Lowest quartile ($\approx 0\text{--}129$)	916	182.24	182.24
Second quartile ($\approx 129\text{--}259$)	913	57.34	56.78
Third quartile ($\approx 259\text{--}380$)	913	70.45	-70.45
Highest quartile ($\approx 380\text{--}500$)	913	190.33	-190.33

Table 10. Prediction Error by Observed AQI Range (Mean Baseline)